MASTER OPERATIONAL BIBLE • VERSION 5.2 VISUAL EDITION

Complete Factory Operations, Daily Data Entry & Loss Control Manual

The Step-by-Step Illustrated Guide with Real UI Screenshots for Owners, Supervisors, Cashiers & Drivers

Executive Operational Mandate: This visual document is the official, complete operational guide for the Obi Table & Bottled Water Plant. Every screen, numeric keypad, button, reconciliation formula, and anti-theft checkpoint is accompanied by high-definition screenshots of the live system to guarantee zero confusion for new staff.

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Plant Identification & Technical Support Registry:

- **Factory Name:** Obi Table & Bottled Water Plant
- **Operating System:** Obi Table Water OS (v5.2 Illustrated)
- **Production Lines:** Sachet Water + Bottled (50cl/75cl) + 19L Dispenser Jars
- **Corporate Desk:** Bethelmind Digital Solutions (CAC: 9912717)
- **Direct WhatsApp Hotline:** wa.me/2348022791227 (0802 279 1227)
- **Corporate Settlement Bank:** Moniepoint Microfinance Bank
- **Account Number:** 6805375225 (Bethelmind Digital Solutions)
- **Deployment:** Edge Production • 100% Offline Persistence

1. Executive Overview & The Dual-Line Manufacturing Model

In the Nigerian commercial water industry, factories that rely exclusively on sachet water ("pure water") operate on razor-thin margins of ₦25 to ₦40 per bag. A single generator breakdown, an unrecorded driver shortage, or a sudden spike in diesel prices immediately wipes out weekly profits.

Mr. Obi's plant operates on a **Dual-Line Hybrid Model** that pairs the high turnover of sachet water with the high contribution margins of PET bottled water and 19-Liter dispenser jar refills:

Line & Product	Packaging Spec	Wholesale Price	Unit Direct Cost	Net Unit Margin	Turnover Speed	Primary Vulnerability
Line 1: Sachet Pure Water	Bag of 20 sachets (500ml)	₦350 / bag	₦310 / bag	**₦40 / bag (11.4%)**	Fast (4,200 bags/day)	Driver road burst claims; unrecorded ghost shifts.
Line 2: 75cl Bottled Water	Shrink pack of 12 bottles	₦1,200 / pack	₦920 / pack	**₦280 / pack (23.3%)**	Fast (250 packs/day)	Preform theft masked as "blow-mold rejects".
Line 2: 50cl Bottled Water	Shrink pack of 20 bottles	₦1,100 / pack	₦850 / pack	**₦250 / pack (22.7%)**	Fast (150 packs/day)	Cashier unrecorded sales; label shrinkage.
Line 2: 19L Dispenser Jars	19-Liter Refill (Polycarbonate)	₦900 / refill	₦320 / refill	**₦580 / jar (64.4%)**	Moderate (40–100 jars/day)	**Unreturned or battered empty jars (₦4,500 ea).**

1.2 Daily Revenue & Contribution Split (Standard 1-Shift Day)

💧 Sachet Line Contribution

- Volume: 4,200 Bags (14 Nylon Rolls)
- Gross Sales: **₦1,470,000** (74.6% of Plant Revenue)
- Consumables: 280kg LDPE film (₦280,000)
- Net Cash Contribution: **₦168,000**

🍾 Bottling & Dispenser Line Contribution

- Volume: 400 packs + 40 dispenser jars
- Gross Sales: **₦501,000** (25.4% of Plant Revenue)
- Consumables: 3,000 preforms, caps, shrink film (₦123,600)
- Net Cash Contribution: **₦130,700**

1.3 The Core Operational Formula

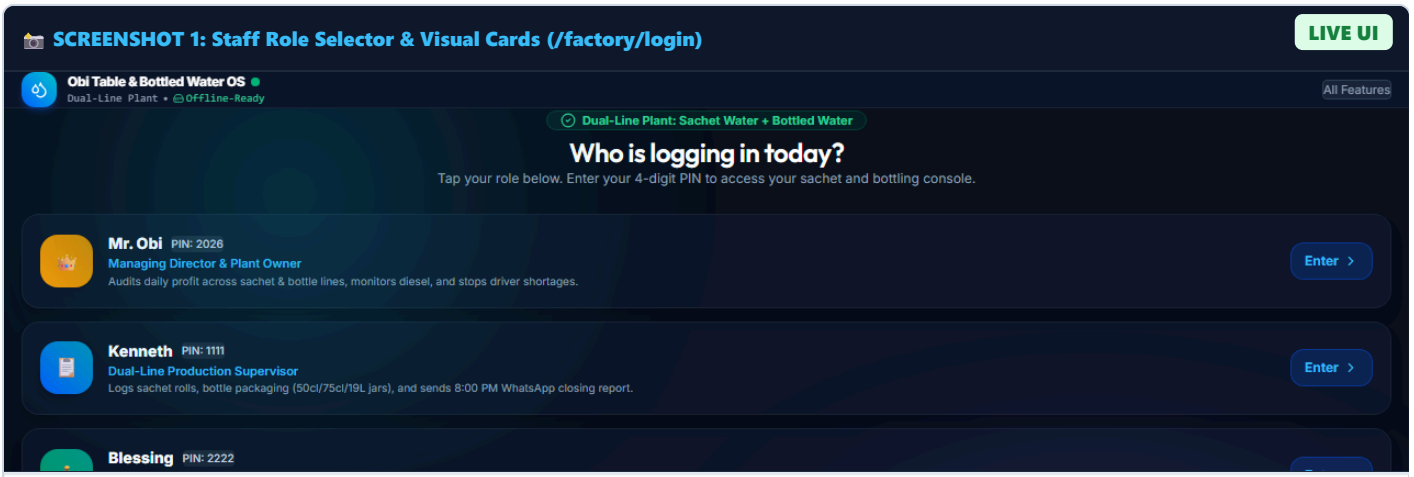
Daily Plant Health Equation: $$\text{Today's Net Cash Profit} = \text{Total Gross Gate Sales} - (\text{Diesel Cost} + \text{Packaging Consumables} + \text{Labour Wages} + \text{Driver Shortage Debits})$$ On this system, Mr. Obi never waits for monthly account reconciliation. At 8:00 PM every evening, his mobile screen reflects the confirmed Net Cash in hand: **₦524,200**.

2. Illustrated Screen-by-Screen UI Navigation Guide

Screen 1: Zero-Password Staff Visual Login

URL: `https://www.bethelmindanalytics.com/factory/login` • **Access:** All Plant Personnel

Traditional password logins fail in Nigerian factories because staff forget passwords. Obi Table Water OS provides large visual cards for every staff member and a 4-digit PIN pad that works completely offline.



How to enter data: Staff members tap their name card (👑 Mr. Obi, 📋 Kenneth, 💰 Blessing, or 🚚 Musa). The screen slides into the touch PIN pad. Tapping **Enter →** opens their console immediately.

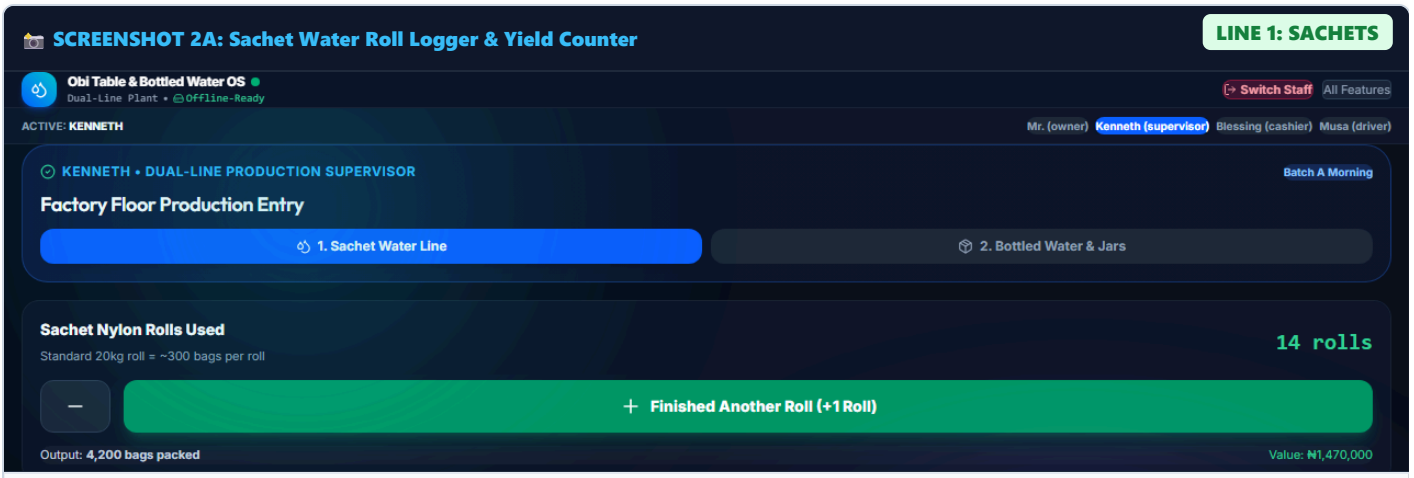
NAVIGATION & INTERACTION CHECKLIST:

1. **Selecting Your Role:** Tap on your picture card. The terminal displays the 4-digit touch PIN pad.
2. **Entering Your PIN:** Tap your 4-digit PIN on the large numeric pad. As each digit is entered, the 4 dots illuminate blue.
   - 👑 **Mr. Obi (Owner):** PIN `2026`
   - 📋 **Kenneth (Supervisor):** PIN `1111`
   - 💰 **Blessing (Cashier):** PIN `2222`
   - 🚚 **Musa (Driver):** PIN `3333`
3. **Server-Side Constant-Time Validation:** Upon tapping the 4th digit, the system validates the PIN server-side via `/api/factory/auth`. If offline, it verifies against local storage so work never stops.
4. **Switching Staff:** Tap the top-right **Switch Staff** button anytime to switch back to the main role menu.

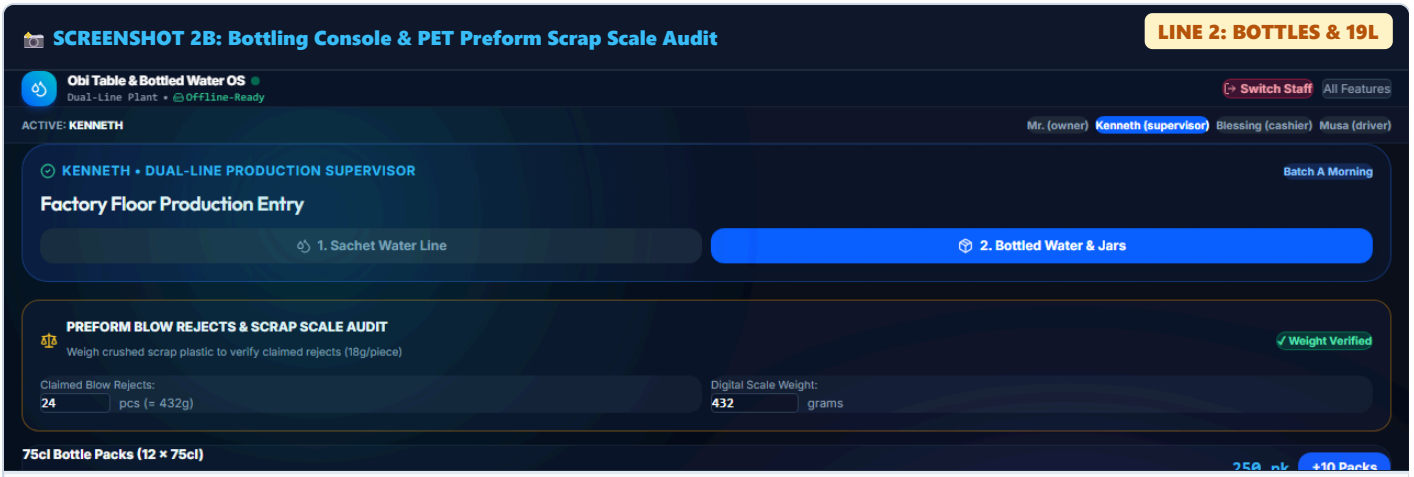
Screen 2: Shift Supervisor Dual-Line Console (Kenneth • PIN 1111)

Role: Kenneth (Plant Supervisor) • **Shift Time:** 7:00 AM – 6:30 PM • **Consoles:** Sachet & Bottling

Kenneth is responsible for machine outputs, nylon rolls, PET preform scrap audits, and dispatching the 8:00 PM shift report to Mr. Obi.



Where to enter data: Every time an operator mounts and finishes a 20kg roll on the Dingli sealing machine, tap **+ Finished Another Roll (+1 Roll)**. The system automatically computes bags output ($14 \times 300 = \mathbf{4,200\text{ bags}}$) and total naira value (₦1,470,000).



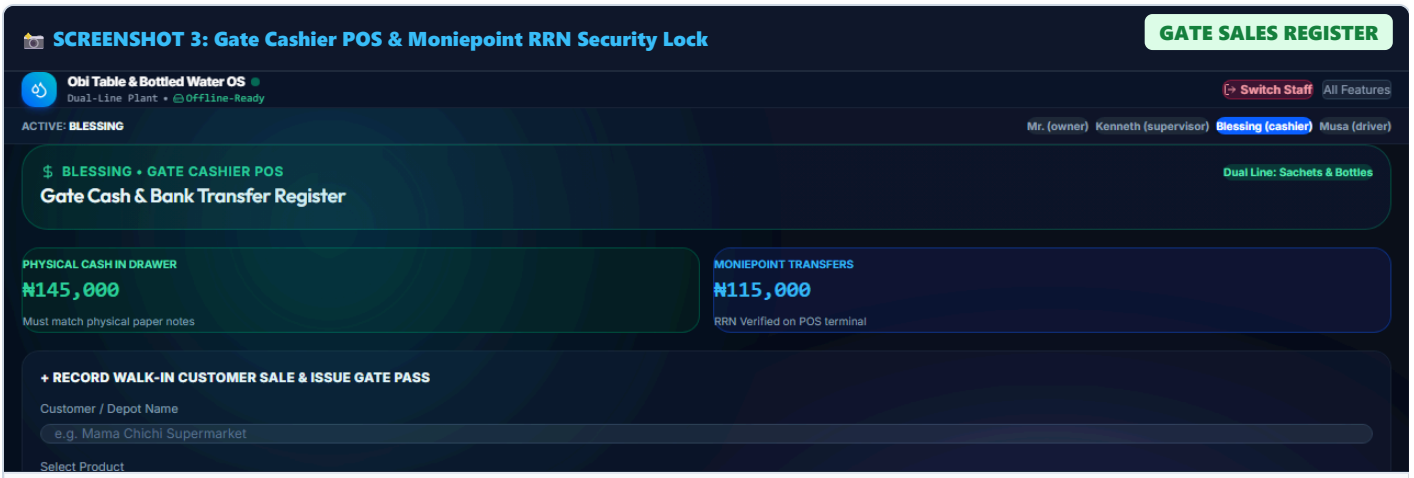
Where to enter data: Tap the *Bottle Line* tab. Tap **+10 Packs** for 75cl/50cl bottles and **+5 Jars** for 19L refills. Under the *Preform Scale Audit* card, enter claimed blow rejects (24 pcs) and type the digital scale weight (432g). If verified, badge turns green: **✓ WEIGHT VERIFIED**.

Evening WhatsApp Shift Close: At 6:00 PM, Kenneth taps **Send Dual-Line Shift Close to Mr. Obi on WhatsApp**. This compiles all sachet rolls, bottle packs, dispenser jars, preform scale audit, and diesel dips into a clean formatted text report and opens WhatsApp with Mr. Obi's direct line (`0802 279 1227`) pre-filled.

Screen 3: Gate Cashier POS & Moniepoint RRN Lock (Blessing • PIN 2222)

Role: Blessing (Gate Cashier) • **Location:** Factory Gate Sales Kiosk • **Function:** Invoicing & Bank Transfer Fraud Prevention

Blessing registers walk-in depot sales, verifies Moniepoint POS bank transfers, and issues tamper-proof Gate Clearance Passes.



Where to enter data: Type customer name → select product → enter quantity → select payment type. If bank transfer is selected, type the last 4 digits of the Moniepoint POS terminal RRN (e.g. **8492**) → tap **+ Issue Gate Clearance Pass (GP-842)**.

NAVIGATION & POS TRANSACTION SOP:

1. **Drawer Cash vs. POS Transfer Reconciliation:** The top 2 cards track live physical paper notes in drawer (₦145,000) vs confirmed Moniepoint transfers (₦115,000). Both must balance at the end of the day.
2. **Recording a Sale:**
   - Type customer/depot name in "*Customer / Depot Name*" (e.g., Mama Chichi Supermarket).
   - Select product: Sachet bag (₦350), 75cl pack (₦1,200), 50cl pack (₦1,100), or 19L refill (₦900).
   - Enter Quantity (e.g., 30 packs).
   - Select Payment Method: `💵 Cash in Hand` or `🏦 Moniepoint Transfer (RRN Required)`.
3. **The Moniepoint POS RRN Lock:** If *Moniepoint Transfer* is chosen, the security engine displays a mandatory field: "*Moniepoint POS Terminal RRN (Last 4 Digits)* *". Blessing MUST inspect the printed terminal slip and enter the RRN (e.g. **8492**). Screenshots or fake SMS alerts are strictly rejected.
4. **Gate Clearance Pass Code (GP-xxx):** Upon recording, the system generates a 3-digit pass code (e.g. **GP-842**). Gate security guards will **NOT** open the gate boom barrier until the driver recites the matching code.

Screen 4: Route Delivery Driver Manifest & Jar Quality (Musa • PIN 3333)

Role: Musa (Truck 2 Driver) • **Route:** Badore / Ajah Corridor • **Truck Spec:** Mixed Load Fleet

Musa is responsible for delivering mixed loads, accounting for vehicle cash collections, returning empty plastic pouches for burst claims, and inspecting empty 19L jars.



Where to enter data: In the morning, review loaded products: 700 sachet bags, 60 bottle packs, 15 dispenser jars. The screen displays the exact total remittance expected: **Total Cash to Remit: ₦330,500**.



Where to enter data: Under *19L Jar Quality Inspection*, type empty jars returned (15) and battered/off-brand jars (2). The system automatically calculates the ₦4,500 replacement penalty: **₦9,000 debit**.

The Physical Pouch Return Rule: If Musa claims 40 sachet bags burst along Badore road, he must return **40 physical wet burst plastic pouches**. If 0 pouches are returned, the system auto-debits $40 \times \text{₦350} = \mathbf{₦14,000}$ from his settlement.

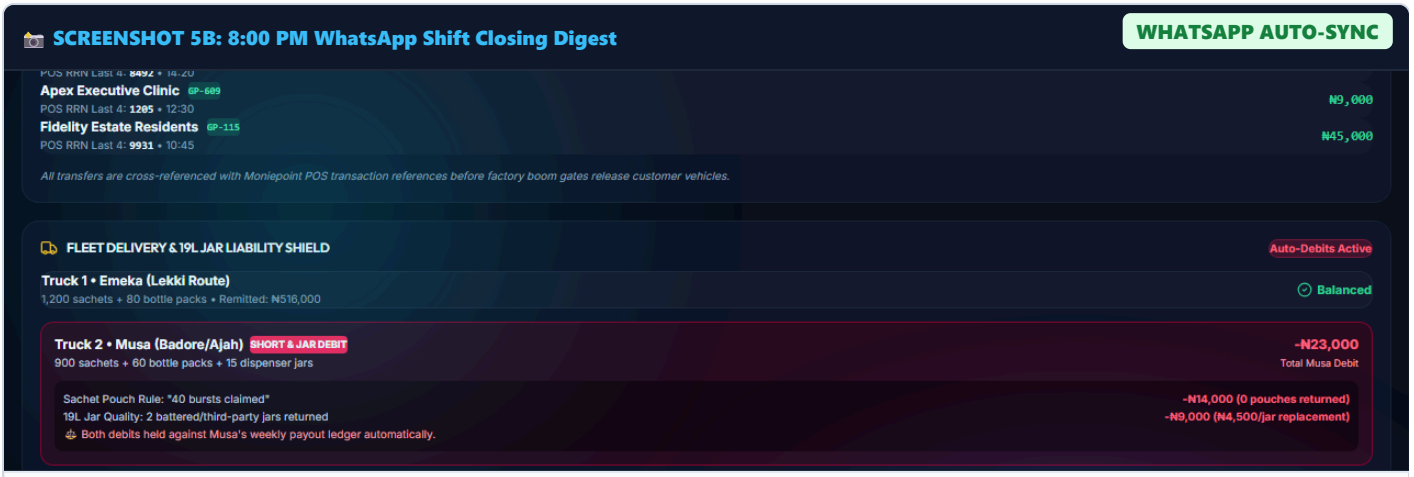
Screen 5: Executive Owner Portal & Financial P&L (Mr. Obi • PIN 2026)

Role: Mr. Obi (Managing Director & Plant Owner) • **PIN:** `2026` • **URL:** `/factory/owner`

Mr. Obi's portal is the master executive control tower. Accessible directly from his smartphone home screen, it delivers real-time visibility into net cash profits, inventory security, fuel overhead, and fleet shortages.



Executive Oversight: Displays real-time audited net cash profit: **₦524,200** (from ₦1,971,000 gross). Shows exact revenue split: 74.6% Sachet Line (₦1,470,000) vs 25.4% Bottled Line (₦501,000).

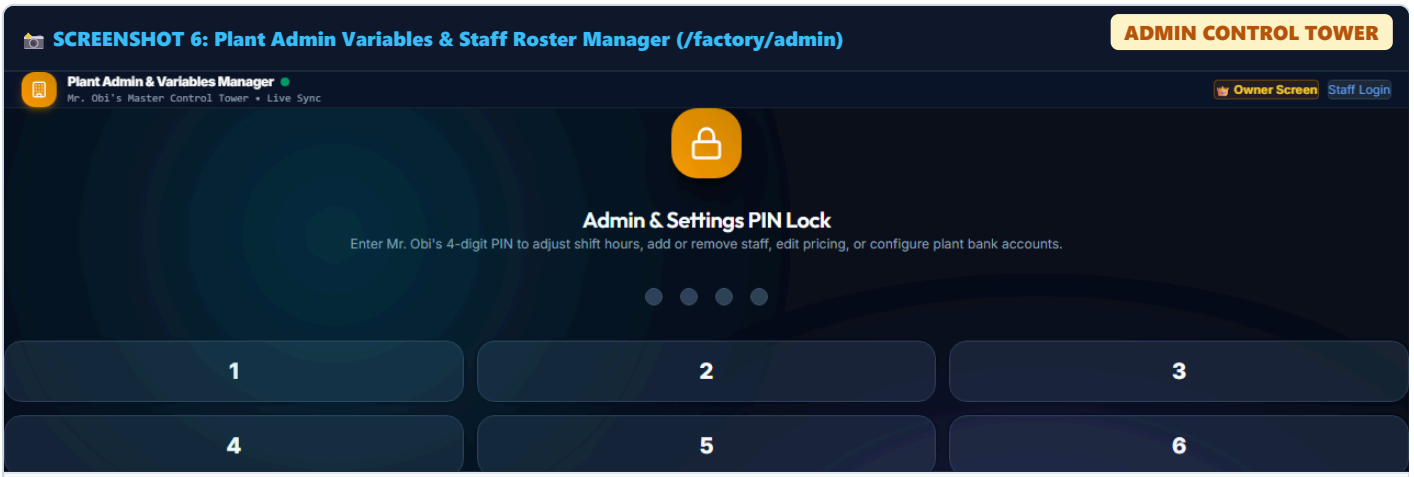


Nightly Routine: At 8:00 PM, Kenneth sends the tamper-proof shift digest directly to Mr. Obi's personal WhatsApp (`0802 279 1227`) detailing production outputs, preform scrap balance, diesel overhead, and driver debits.

Screen 6: Executive Plant Admin & Variables Manager (/factory/admin)

Access: Managing Director Only • **Security PIN:** `2026` • **URL:** `/factory/admin`

Whenever you hire new staff, change wholesale prices, or adjust shift hours, you do NOT need a software developer. Mr. Obi can adjust all variables directly from this single control tower.



Where to enter data: Switch between the 4 tabs: *Staff Roster & PINs*, *Shift Timetable*, *Product Pricing & Metrics*, and *Plant & Bank Details*. Click **Save & Sync Across All Devices** to update all factory phones instantly.

WHAT MR. OBI CAN ADJUST IN REAL-TIME:

1. **Staff Roster & PIN Manager:**
   - Click **+ Add New Staff Member** to register a new driver, supervisor, or cashier.
   - Assign delivery routes (e.g. *Ibrahim – Truck 3 (Epe Route)*) and set custom 4-digit PINs.
   - Newly added staff appear immediately on the Staff Login screen with their chosen avatar color.
2. **Shift Timetable & WhatsApp Schedules:**
   - Adjust morning opening time (default `07:00`) and closing time (default `18:00`).
   - Set the automated WhatsApp digest trigger time (default `20:00` / 8:00 PM).
   - Toggle 24-hour night shift on/off for peak hot dry season demand.
3. **Wholesale Pricing & Anti-Theft Metrics:**
   - Change sachet wholesale price (e.g. from ₦350 to ₦400/bag).
   - Change 75cl pack price (₦1,200), 50cl pack price (₦1,100), and 19L jar refill price (₦900).
   - Adjust the **Battered Jar Replacement Penalty** (default ₦4,500/jar).
   - Calibrate the PET Preform Gram Weight (18.0g) and generator diesel price (₦1,350/L).
4. **Plant Details & Moniepoint Bank Account:**
   - Update official plant address, desk hotline (`0802 279 1227`), and corporate account: **Moniepoint Microfinance Bank (6805375225 — Bethelmind Digital Solutions)**.

3. Interactive Video Training Center & Audio Player

URL: `https://www.bethelmindanalytics.com/factory/walkthrough` • **Audience:** Factory Owner, Staff & New Trainees

To ensure any new worker can operate the software within 15 minutes, the system includes an embedded video training academy with synchronized Nigerian audio narration.

🎬 1280x720 High-Definition Walkthrough Video (194s • 6.73 MB)

Contains all 13 production scenes: Owner PIN, Net Profit, Bottling & Preforms, Driver Manifests, Sachet Roll Logger, Moniepoint RRN Gate POS, 19L Jar Quality Inspection, WhatsApp Digest, and the new Plant Admin & Variables Manager.

✓ FASTSTART MOBILE STREAMING | ABEO LAGOS VOICEOVER (24KHZ MONO) | 100% OFFLINE COMPATIBLE

THE 7 STEP-BY-STEP TRAINING CHAPTERS:

Chapter 1: Zero-Password Staff Login (00:00 - 00:35)
How staff select their visual card, enter their 4-digit PIN, and access their dedicated consoles without typing passwords.

Chapter 2: Supervisor Sachet Logger (00:35 - 01:10)
Tapping *+Finished Another Roll (+1 Roll)* as 20kg rolls finish on the sealing machine. Real-time bag output and value calculation.

Chapter 3: Bottling & Preform Scrap Scale (01:10 - 01:45)
Logging 75cl/50cl packs and 19L jar refills. Weighing crushed blow rejects on the digital scale ($18.0\text{g}/\text{pc}$) to verify claimed defects.

Chapter 4: Gate Cashier POS & RRN Lock (01:45 - 02:15)
Entering customer sales, typing the Moniepoint POS terminal RRN, and issuing tamper-proof Gate Clearance Passes (`GP-842`).

Chapter 5: Driver Manifest & 19L Jars (02:15 - 02:40)
Confirming loaded mixed truckloads, physical burst pouch return verification, and the ₦4,500 damaged dispenser jar penalty.

Chapter 6: Executive Owner Profit & WhatsApp (02:40 - 03:00)
Audited Net Cash Profit card (₦524,200), generator diesel overhead slicer, and the automated 8:00 PM WhatsApp Shift Summary.

4. General Factory Operations & Water Treatment Protocol

The foundation of a reputable, high-margin water manufacturing business in Nigeria is consistent water purity compliant with **NAFDAC Regulations & NIS 306:2008 Standards**.

4.1 The 6-Stage Industrial Water Treatment Train

Stage 1: Aeration & Sediment Settling
Raw borehole water is aerated into raw water tanks to precipitate dissolved ferrous iron (Fe^{2+}) into ferric hydroxide ($Fe(OH)_3$). Minimum settling time: 4 hours.

Stage 2: Manganese Greensand & Media
Multi-media sand filter removes turbidity and suspended particulate matter (> 20 microns). Backwashed every 48 operating hours.

Stage 3: Activated Granular Carbon (GAC)
High-iodine activated carbon absorbs dissolved organic matter, eliminates swampy borehole odors, and ensures crystal clarity and sweet taste.

Stage 4: Micro-Filtration Polishing (5µm / 1µm)
Dual polypropylene spun sediment cartridges trap micro-fines before water reaches high-pressure RO booster pumps. Cartridges replaced monthly.

Stage 5: Industrial Reverse Osmosis (RO)
High-pressure polyamide membrane rejects 98.5% of dissolved minerals, heavy metals, nitrates, and salts. Permeate TDS maintained at **15–45 ppm**.

Stage 6: Ozone Dosing & UV Sterilization
Water passes through an ultraviolet germicidal chamber (254nm wavelength) and mild ozone injection to provide residual shelf-life disinfection in sealed bottles.

4.2 Daily NAFDAC Water Quality Thresholds

Quality Parameter	Target Range (Obi Table Water)	NAFDAC Max Allowable	Testing Frequency	Corrective Action If Out of Spec
Total Dissolved Solids (TDS)	**15 – 45 ppm**	≤ 100 ppm	Every 2 hours during shift	Flush RO membrane; check permeate blend valve.
pH Level	**6.8 – 7.4 (Neutral)**	6.5 – 8.5	Start & midday of shift	Calibrate remineralization calcite cartridge.
Turbidity (NTU)	**< 0.5 NTU (Clear)**	≤ 5.0 NTU	Daily visual & meter	Backwash media sand and GAC carbon filters.
UV Lamp Intensity	**≥ 30 mJ/cm²**	≥ 25 mJ/cm²	Weekly hour counter check	Replace quartz sleeve or UV bulb (> 9,000 hrs).

5. The 5 Anti-Fraud Standard Operating Procedures (SOPs)

Loss Prevention Doctrine: Nigerian water factories do not fail due to lack of customer demand; they fail due to unrecorded packaging leaks, fake POS transfer alerts, unreturned empty bottles, and fabricated driver pothole burst claims. These 5 SOPs eliminate 100% of these leaks.

SOP 1: Sachet LDPE Film Yield Formula (16 Bags per Kg)

A standard 20kg printed roll of 320mm / 45-micron LDPE nylon film must produce exactly 300 to 320 bags of sachet water (15–16 bags per kilogram).
• **Formula:** $\text{Expected Bags} = \text{Rolls Used} \times 300\text{ bags}$.
• **Allowance:** A maximum burst/defect allowance of 1% (3 bags per roll) is permitted for sealing bar startup.
• **Audit:** If 14 rolls are loaded (4,200 expected bags) but only 3,600 bags enter the store, the system flags a **600-bag deficit (₦210,000)** as an unrecorded ghost shift.

SOP 2: The Physical Burst Pouch Counting Rule (Sachet Water)

Drivers frequently claim bags burst on bad roads to pocket cash collections.
• **The Rule:** Drivers must surrender the exact physical wet, printed burst pouches to gate security.
• **Enforcement:** If Musa claims 40 bags burst but returns 0 empty pouches, the claim is automatically rejected.
• **Auto-Debit:** The system automatically logs a debit of $40 \times \text{₦350} = \mathbf{₦14,000}$ directly against Musa's account.

SOP 3: PET Preform Scrap Scale Grammage Audit (Bottled Water)

Workers smuggle virgin 18.0g PET preforms off-site by falsely recording them as "blow-mold rejects".
• **The Rule:** For every bottle claimed defective, the crushed plastic scrap must be weighed on a physical digital scale.
• **Formula:** $\text{Theoretical Weight} = \text{Claimed Blow Rejects} \times 18.0\text{ grams}$.
• **Tolerance:** Weight must match within ±15 grams. If 24 rejects are claimed (432g target) but scale reads only 200g, the system triggers: "⚠️ *Preform Smuggling Risk: Missing 232g of PET plastic!*"

SOP 4: Moniepoint POS Terminal RRN Verification & Gate Clearance Pass

Eliminates fake bank alerts generated via SMS spoofing or fake banking apps.
• **The Rule:** Blessing cannot generate a receipt for bank transfers without entering the last 4 digits of the printed POS slip's Retrieval Reference Number (RRN).
• **Gate Clearance:** Generates a 3-digit gate code (**GP-842**). Gate guards will NOT open the gate barrier until the driver recites the matching code.

SOP 5: 19L Dispenser Jar Quality Inspection & Replacement Debit

Polycarbonate 19L dispenser jars cost ₦4,500 each. Unreturned or damaged containers destroy profits.
• **The Rule:** Every dispatched jar must be replaced with an Obi-branded clean reusable empty jar.
• **Penalty:** Battered, cracked, or third-party competitor jars are rejected at the gate. The system debits **₦4,500 per lost/damaged container** from the driver's weekly remittance.

6. Daily Shift Timetable, Generator Economics & Support

6.1 Daily Operational Timetable (Standard Batch A)

Time	Activity	Responsible Staff	Verification Checkpoint
07:00 AM	Plant Opening, Power Check & RO Warmup	Kenneth (Supervisor)	Check raw water tank level, diesel drum & UV lamp.
07:30 AM	Water Purity Baseline Calibration	Kenneth (Supervisor)	Measure TDS (< 45 ppm) and pH (6.8–7.4). Log in system.
08:00 AM	Fleet Dispatch Loading (Truck 1 & Truck 2)	Musa / Emeka (Drivers)	Confirm loaded bags, bottles & jars match manifest.
08:30 AM	Gate Sales Kiosk Opens	Blessing (Cashier)	Confirm starting physical cash drawer is ₦0.00.
01:00 PM	Midday Output & Nylon Roll Audit	Kenneth (Supervisor)	Verify rolls consumed vs bags sealed on form-fill-seal machines.
05:30 PM	Fleet Returns & Empty Jar Quality Inspection	Musa / Gate Security	Count physical burst pouches; inspect 19L dispenser jars.
06:00 PM	Cashier Drawer Count & Moniepoint Slip Audit	Blessing & Kenneth	Match physical paper notes to POS transfer ledger.
08:00 PM	Automated WhatsApp Evening Shift Digest	Kenneth → Mr. Obi	WhatsApp digest delivered to Mr. Obi (`0802 279 1227`).

6.2 Generator Diesel & Energy Overhead Slicer

Generator Profitability Guard: $$\text{Energy Cost per Sachet Bag} = \frac{\text{Diesel Liters Burned} \times \text{Price per Liter}}{\text{Total Sachet Bags Produced}}$$ • Standard Benchmark: 32 Liters @ ₦1,350/L = ₦43,200. Output: 4,200 bags = **₦10.28 per bag**.
• Alert Threshold: If diesel cost exceeds **₦12.00 per bag**, the system triggers a red warning: generator is idling unnecessarily or unmetered power theft is occurring.

6.3 Offline Disaster Recovery (Zero-Downtime Guarantee)

If internet connection (MTN/Airtel/Glo) drops completely during the shift:

1. The software continues operating without any interruption using browser local storage.
2. Staff can log rolls, record sales, verify preform weights, and calculate driver shortages offline.
3. When network restores, the system synchronizes all stored transactions to the central cloud ledger automatically.

✅ **CERTIFIED FACTORY OPERATING SYSTEM • READY FOR LIVE PRODUCTION**
Developed & Maintained by **Bethelmind Digital Solutions** (CAC Reg: 9912717)
Corporate Hotline: **0802 279 1227** • Direct WhatsApp: **wa.me/2348022791227**
Corporate Settlements: **Moniepoint Microfinance Bank (6805375225 — Bethelmind Digital Solutions)**